Cryptocurrency Market Dynamics Around Ethereum's Merge: Intraday Volatility, Volume and Liquidity

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Notebooks, sources and data are available on GitHub

1 Introduction

Nakamoto (2008) introduced Bitcoin as a mechanism enabling peer-to-peer decentralized electronic payments, without the need to rely on financial institutions for transaction validation. In practice, a network of independent nodes – computers running the protocol – maintains a record of ownership and all transfers of value by organizing transactions into blocks identified by a hash. To function, validators – the nodes responsible for verifying blocks – must reach consensus on a unique version of this ledger. Consensus can only be achieved if nodes follow a given sequence of actions needed to maintain the blockchain: the protocol (Amoussou-Guenou et al., 2026).

Validators play a critical role in order for this system to function without any central authority: they must be incentivized to fulfill their role, and to do so while ensuring compliance with the protocol. With no global oversight, incentive compatibility – a combination of rewards (newly issued tokens distributed to validators) and penalties (loss of computing resources or staked capital) – fosters alignment. In Bitcoin's Proof-of-Work (PoW) consensus mechanism, anyone can become a validator to take part in the competition among nodes, by trying to propose the next block by solving computation-intensive puzzles, with winning odds being weighted by computational power. Winners are rewarded, while invalid blocks are rejected, each error sinking computational effort and energy expenditure. Biais et al. (2019) show that PoW constitutes an equilibrium, although it is not the only one: Amoussou-Guenou et al. (2024) obtain a reciprocal result for committee-based Proof-of-Stake (PoS) protocols.

Ethereum extends Bitcoin's decentralization beyond payment with smart contracts: self-executing programs allowing for a broader set of applications in decentralized finance. On September 15th, 2022, the Ethereum network underwent the Merge: its PoW layer merged with a parallel PoS layer that had been running since 2020. Unlike in PoW, PoS validator status depends on meeting a threshold of staked capital, with next-block validating node selection being weighted by stake. Validators face the risk of losing their capital if blocks they propose deviate from the protocol. This transition from intensive block "mining" to staked validation cut the network's energy consumption by 99.95% and brought Ethereum one step closer to its roadmap setting the stage for later scalability upgrades while leaving transaction history and user holdings intact. Since these consensus mechanisms rely on specific reward and penalty structures, a shift in protocol could alter this balance of incentives, and by extension impact the financial markets on which these assets are traded.

The Merge therefore offers a rare natural experiment: a structural change in the protocol of the second largest cryptocurrency by market capitalization. This setting allows us to study how a protocol shift affects the trading behavior of a cryptocurrency, specifically the dynamics of volatility, volume and

liquidity, whereas Bitcoin's unchanged protocol serves as a neutral counterfactual. To capture the effects of the Merge, our observation window spans from May 4th, 2021 to December 31st, 2024. This choice stems from data availability constraints and aims to maintain consistent market conditions throughout our sample in the fast-moving landscape of cryptocurrencies: prior data originate from a vastly different market environment, characterized by incomparable volumes and dynamics.

We find that the Merge itself had little observable effect on Ethereum's trading dynamics across volatility, liquidity and volume measures. Cointegration between Ethereum and Bitcoin persists throughout the Merge, with no clear break in their long-run relation, contrary to what a significant shift of Ethereum metrics caused by the Merge would imply. By comparison, a simultaneous and unrelated exchange-level fee change for Bitcoin pairs listed on Binance – resulting in a structural shift of similar scope for trading metrics, confined to Binance's Bitcoin pairs – had a more pronounced impact on Bitcoin's trading volume than the Merge had on Ethereum's. Finally, our extended HAR-RV regressions show no shift in the regressors driving Ethereum's volatility around the Merge, indicating that the mechanisms linked to this metric remained unchanged by the transition. These results suggest that the Merge had no significant impact on Ethereum's spot market behavior.

Literature shows that empirical evidence of the Merge's effect on markets remains limited and mixed. Tsuyuguchi and Wang (2025) use a DiD design to capture ETH-BTC dynamics: they find that daily returns and on-chain fees were not significantly affected by the Merge, while trading volumes and network activity diverged significantly post-transition. Liu et al. (2025) report that the change was translated into improved liquidity, reduced trading activity and a decrease in intraday volatility for Ethereum relative to Bitcoin. These diverging conclusions emerging from different empirical strategies motivate a closer look at the volatility-volume-liquidity relationship itself. Jain et al. (2023) studies the impact of the Merge on on-chain fees, showing that the transition reduced them for both Bitcoin and Ethereum blockchains. This mechanism is distinct from the exchange-level fee regime changes we observe on Binance, but is relevant to distinguish effects of protocol-level from exchange-level changes on trading dynamics. Pertinent to the horizons central to our HAR-RV approach, Hansen et al. (2024) confirms the intraday and weekly periodicity patterns in Bitcoin and Ethereum volume we observe in our data. Finally, Bogousslavsky and Collin-Dufresne (2023) equity-market perspective links order imbalance volatility to the volume-spread relationship. Although based on equity, their framework motivates the volume-illiquidity channel we test empirically through use of Amihud, Realized Amihud (Ranaldo and de Magistris, 2022) and Kyle's lambda.

This report is organized as follows. Section 2 describes data sources for trading data and events used throughout our analysis. Section 3 introduces the intraday volatility and illiquidity estimators we use, and justifies our bin size choice. Section 4 documents market periods, exchange-level fee changes, cointegration testing across periods, flight of capital from PoW to PoS assets, and estimation of HAR-RV models linking volatility to volume, liquidity and fees. Finally, Section 5 synthesizes our findings and discusses possible directions for further causal analysis.

2 Data

2.1 Market Data

Binance, the world's largest cryptocurrency exchange by trading volume, provides an exceptionally dense and well-documented public dataset for all of its trading pairs. For spot, it includes tick-by-tick trade history and candlestick data down to a 1s resolution, ranging from 2017-08 to 2026-07. For our observation window, we store aggregated trades (3.7B rows) and 1s price data (231M rows) into a DuckDB database for Ethereum-Tether (ETHUSDT) and Bitcoin-Tether (BTCUSDT) pairs. The choice of Tether pairs is justified by the market capitalization of the USDT stablecoin – ranked third behind Bitcoin and Ethereum – and the substantial historical data available for its pairs on Binance.

In theory, trade data allows us to reconstruct higher bin sizes by aggregating trades over intervals of duration N. The effectiveness of this approach depends on the fidelity of raw trade feeds. In practice, a 100% Pearson correlation and a zero median discrepancy between downloaded and reconstructed 1s data confirm the reliability of this method, that we use for later bin size reconstruction.

Unlike traditional financial markets where prices are centralized, the cryptocurrency system is characterized by its fragmentation across multiple exchanges of varying volumes, listing closely aligned prices that would otherwise offer arbitrage opportunities (Makarov and Schoar, 2020 Crépellière et al., 2023). For cross-validation of our Binance dataset, we use Coinbase, the second largest exchange by volume. Its API provides 1m historical candlestick data for Bitcoin and Ethereum, respectively spanning from 2021-05 and 2017-07 to 2026-05. Coinbase's candlesticks, quoted in either USDT or USD, are seemingly interchangeable, as underlined by a near 100% Pearson correlation and a median absolute price difference situated below 0.05% (below 0.3% at the 99th percentile). However, their trading activity differs significantly: USD volumes are 25 to 35 times higher than those of USDT. We therefore select the former for subsequent inter-exchange comparisons. A Binance-Coinbase comparison for both cryptocurrencies reveals a near 100% Pearson correlation and a median price difference below 0.02% for candlestick data. Fewer than 0.01% of 1m bins closing prices exhibit a discrepancy greater than 1%, emphasizing limited upfront inter-exchange arbitrage opportunities. Given that Binance's USDT pairs have 78 to 128 times the volume of equivalent Coinbase USD contracts, we retain the former as our default data source. Other major exchanges (Kraken, Bybit) were evaluated as alternative sources, but either lacked directly comparable trading pairs or provided limited historical coverage.

2.2 Market Events

To provide context for our empirical analysis, we compile a timeline of significant events that affected Ethereum and Bitcoin markets in the period surrounding the Merge. For Ethereum, sources include Ethereum Project's forks timeline that lists major milestones of the blockchain, Ethereum Foundation's blog and ETH101's timeline discussing exogenous events affecting the crypto environment during Ethereum's lifespan. CoinMarketCap's all-time Ethereum and Bitcoin plots listing news and milestones, Binance archived announcements for fees regime, as well as Strategy's Bitcoin purchases were used for Bitcoin. Events span two main categories: Ethereum's protocol upgrades (e.g. the Merge), and global market shocks including regulatory changes. Across hundreds of events, those listed in Appendix A.1 pinpoint major volume-shifting events for these markets over our timeframe. To control for financial context, we obtain 5m interval S&P 500 OHLC data from a Kaggle dataset ranging from 2019-11 to 2024-08, and weekly risk-free rate from Kenneth R. French's website ranging from July 1926 to June 2026.

3 Metrics

Intraday measures of volatility and liquidity are derived from aggregated trade data grouped at frequencies ranging from 10ms to 1h bins, subsequently aggregated at a daily level. For returns computation, we choose a close-to-close measure over an open-to-close alternative, since our data emphasizes that the latter doesn't always account for price movements occurring between the close of one bin and the open of the next. Considering the size of our dataset, loading all trades at once would exceed our physical RAM limitations. We therefore use a day-to-day data streaming approach in order to stay within our capacity bounds when computing metrics. This process is slow – taking over an hour – but allows us to obtain daily aggregated metrics based on various bin sizes.

Intraday volatility is not directly observable: only prices sampled at a given frequency $\Delta\tau$ can be. For day t divided into M intervals $t = t_0 < t_1 < ... < t_M$, let r_{t_i} denote intraday logarithmic returns over the i-th interval (Dumitrescu, 2025). We consider several estimators for volatility. First, realized variance (RV_t,

Eq. 1) is a benchmark estimator of daily variance that converges toward integrated volatility in the absence of jumps, as $M \to \infty$. For a jump-robust alternative, we implement the bipower variation ($BV_t^{\{1,1\}}$, Eq. 2) introduced by Barndorff-Nielsen and Shephard (2004). Realized semi-variance (RS^+, RS^-, Eq. 3) captures variance asymmetry based on signed returns to distinguish the respective contributions of upside and downside risk to total volatility (Barndorff-Nielsen et al., 2010). Finally, to leverage full candlestick data rather than log-returns alone, we include the Garman-Klass estimator ($G\text{-}K_t$, Eq. 4) as an efficient zero-drift variance measure (Garman and Klass, 1980), alongside the Rogers-Satchell alternative ($R\text{-}S_t$, Eq. 5) which provides an unbiased variance estimate in case of price trends (Rogers and Satchell, 1991). No estimator is without flaws and literature identifies the shortcomings of existing measures and justifies alternatives accordingly. Here, RV_t doesn't account fairly for volatility in the presence of jumps, and all realized metrics discard part of the price information carried in candlestick data, while $G - K_t$ and $R - S_t$ rely on assumptions that may not hold empirically. Consistent with the exploratory nature of this analysis, we report all estimates instead of a unique choice that wouldn't be empirically justified in order to cross-validate results across bin size and to explore eventual discrepancies across measures as entry points for further analysis.

$$RV_t = \sum_{i=1}^{M} r_{t_i}^2 \tag{1}$$

$$BV_t^{\{1,1\}} = BV_T = \frac{\pi}{2} \sum_{j=2}^{M} |r_{t_j}| \cdot |r_{t_{j-1}}| \tag{2}$$

$$RS_t^+ = \sum_{i=1}^{M} r_{t_i}^2 \mathbb{1}_{rt_i \geq 0} \quad \text{and} \quad RS_t^- = \sum_{i=1}^{M} r_{t_i}^2 \mathbb{1}_{rt_i \leq 0} \tag{3}$$

$$G\text{-}K_t = 0.5 \left[\ln \left(\frac{H_t}{L_t} \right) \right]^2 - (2 \ln 2 - 1) \left[\ln \left(\frac{C_t}{O_t} \right) \right]^2 \tag{4}$$

$$R\text{-}S_t = \ln \left(\frac{H_t}{C_t} \right) \ln \left(\frac{H_t}{O_t} \right) + \ln \left(\frac{L_t}{C_t} \right) \ln \left(\frac{L_t}{O_t} \right) \tag{5}$$

where O_t, H_t, L_t, C_t represent open, high, low and close values for bin t, respectively.

For illiquidity measures, Amihud's illiquidity ratio (Eq. 6) is the standard reference, measuring illiquidity as a ratio of absolute percentage return to dollar volume over a given period (Amihud, 2002). Its main appeal is its simplicity, requiring only price and volume data. Ranaldo and de Magistris (2022) recently introduced an alternative version of this ratio: the realized Amihud (Eq. 7), which aggregates absolute intraday log-returns before dividing by daily volume in order to capture intraday variation that the Amihud ratio would otherwise average out. Comparing both measures lets us assess whether they converge and whether this added granularity introduces meaningful signal. Finally, Kyle's lambda (Eq. 8) offers an alternative price-impact measure derived from the linear relationship between trade-level order flow and price change (Kyle, 1985). Unlike the previous two illiquidity estimators, this metric can only be computed from the tick-level data our dataset provides, as it requires trade-by-trade data to obtain signed volume.

$$Amihud_t = \frac{|R_t|}{V_t} \tag{6}$$

$$RAmihud_t = \frac{\sum_{i=1}^{M} |r_{t_i}|}{V_t} \tag{7}$$

$$\Delta p_{t_i} = \lambda_t \cdot x_{t_i} + \varepsilon_{t_i} \tag{8}$$

where R_t is daily return, V_t is the daily dollar volume, Δp_{t_i} is the price change associated with trade i,

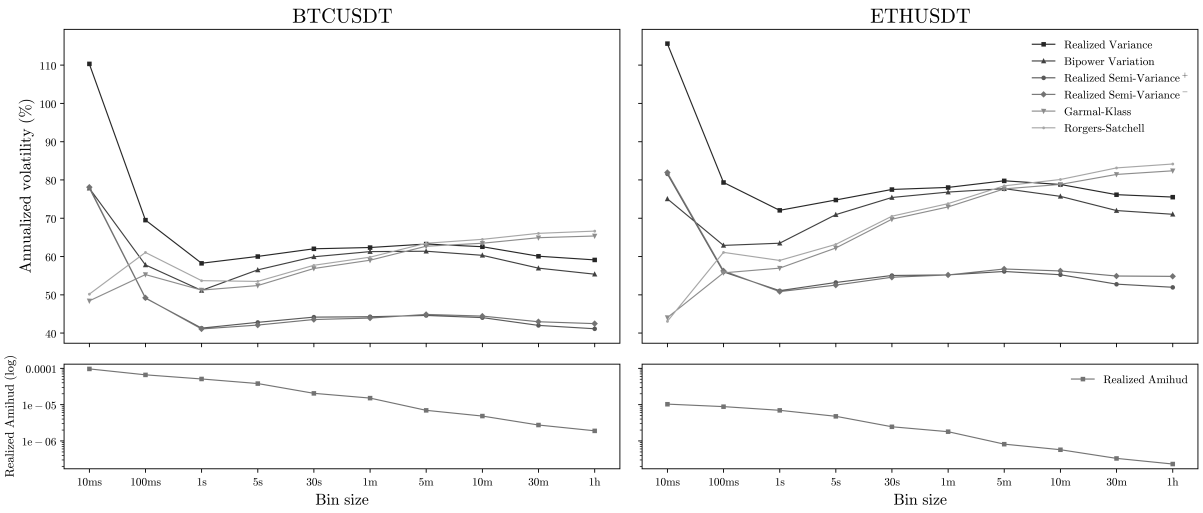
x_{t_i} is the signed trade volume and λ_t measures the price impact per unit of order flow.

Signature plots allow us to observe the evolution of our metrics as bin size increases from 10ms to 1h (Fig. 1). Microstructure noise is noticeable at the 10ms and 100ms intervals for return-based volatility metrics, while Garman-Klass and Rogers-Satchell are less affected by sampling size at these frequencies. A convergence appears from 1s onwards, with all volatility metrics giving similar readings at the 5m interval. This suggests that choice of estimator matters little at such granularity, as they yield comparable results. This high correlation is confirmed by our heatmap analysis (Appendix A.2). For sided metrics, realized semi-variance shows no pronounced asymmetry, consistent with rather balanced directional price movements. We therefore select 5m as the default bin size for subsequent plots and analysis, as a trade-off between fine granularity and microstructure noise. This follows Andersen et al. (2001) who argues that for a liquid market, a 5m sampling interval is optimal, a choice followed by Bandi and Russell (2006), Andersen et al. (2007), Patton (2011) as well as Haugom et al. (2014).

For liquidity metrics, Amihud ratio is computed on a daily basis and Kyle's lambda is obtained through trade-level data: those metrics hence remain unaffected by bin size. Only realized Amihud shows a clear decreasing trend as bin size increases, represented in the bottom subplot of Fig. 1. This result follows an economic intuition: increasing bin size mechanically increases volume per bin, hence reducing the market impact of normal-sized orders on markets where daily volume is measured in billions.

With estimators defined and bin size established, we now proceed to explore market dynamics associated with the Merge.

FIGURE 1: Signature plot of realized volatility and liquidity metrics (10ms to 1h)



4 Analysis and Hypothesis Testing

4.1 Market Dynamics

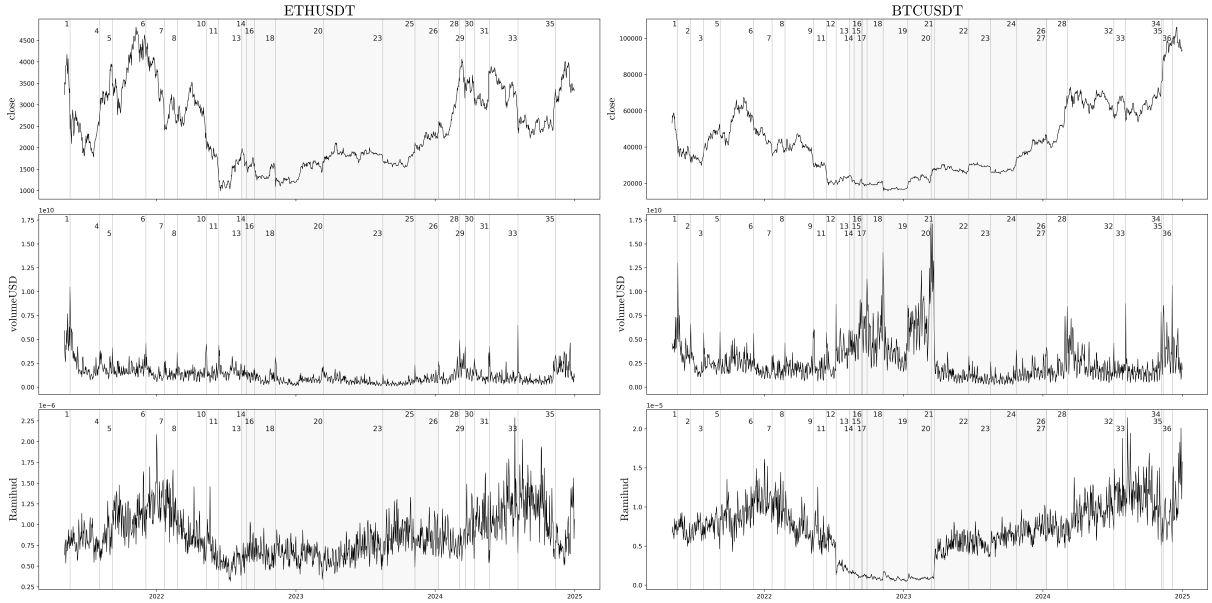
4.1.1 Events and Market Periods

Using 5m data, we plot daily price and dollar volume of Ethereum and Bitcoin pairs alongside all events explaining major volume spikes for these markets. We also add Binance's fee regime changes and Merge-related announcements to the chart. Figure 2 underlines a two-fold difference in volume (average daily volume is \$1.2B for Ethereum, \$2.7B for Bitcoin) and a large difference in asset price. Since a higher realized Amihud value indicates a less liquid market, Ethereum is relatively more liquid than Bitcoin. From a visual reading, Merge announcements (events 13, 14) are linked to little volume change, and

Merge execution (event 16) is linked to a small-scale volume spike for Ethereum. Merge execution has no observable effect on Bitcoin charts compared to, for example, *US inflation announcement higher than forecasted* (event 15). However, Binance's removal then reintroduction of fees on Bitcoin pairs (respectively events 12 and 21), has a notable impact on metrics. For Bitcoin, it is directly linked to a phase of sustained increased volume (+160% compared to the overall fee-present sample trading volume) as trading becomes frictionless. For Ethereum, a mean -26.4% decrease in volume during the fee-less phase indicates that no similar effect is to be observed. This effect is even more visible on realized Amihud, which flattens out throughout the fee-less phase, indicating that increased volume led to improved Bitcoin liquidity. This minute reaction to the Merge and a noticeable response to a trivial fee change are worth addressing in Section 4.2, in order to verify whether volume shifts induced by exchange-fee-regime impact volatility measures.

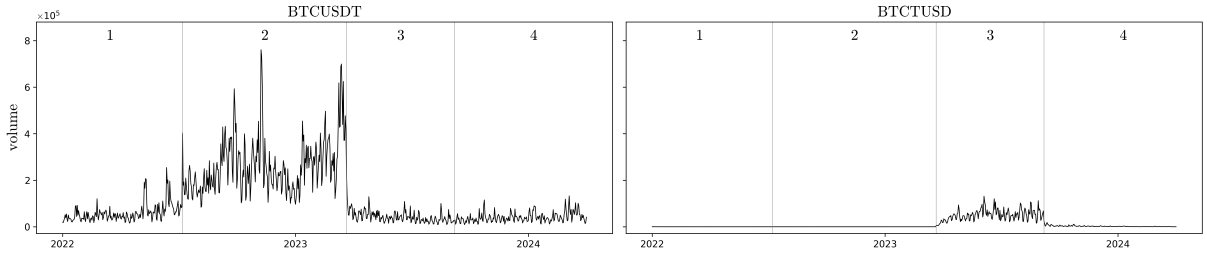
Figure 3 illustrates this fee-related volume shift. Phase 1 corresponds to normal trading fees on Binance (0.1% of the transaction value); phase 2 begins when Binance removed fees on 13 Bitcoin pairs; phase 3 starts when this zero-fee promotion is withdrawn from all pairs except BTCTUSD; phase 4 represents the full reintroduction of fees. BTCUSDT volume rises sharply during phase 2 and drops immediately once fees are reinstated, underlining how fee regime impacts market behavior. BTCTUSD shows a volume spike once it becomes the only fee-less Bitcoin pair listed on Binance. Coinbase volume analysis shows a stable trend throughout these periods, underlining how these volume changes are linked to Binance's promotional-stunt rather than global market context; it is omitted from the figure due to its different scale. During our entire sample, Binance fees for the ETHUSDT pair remained stable at around 0.1%, hence no similar volume shift is observed for this cryptocurrency on this exchange.

FIGURE 2: Daily price, dollar volume and realized Amihud along major market events



Plotting these events allows us to define three market periods shared by both assets, indicated by the grey shading in Figure 2. Period 1 (450 days) follows the DeFi summer and is characterized by increased volatility, as prices reach all-time-highs before plummeting, followed by coin collapses (Terra/LUNA, UST) and multiple crypto lenders and exchanges (Celsius Network, Voyager Digital) filing for bankruptcy. Period 2 (from event 13 to event 26, 516 days) begins with the first blog announcement of the Merge; it is more stable and characterized by regulatory changes as the market moves more horizontally. Finally, the approval of Bitcoin spot ETFs marks the beginning of period 3 (356 days), during which institutional investors gain access to a more volatile market influenced by the outcome of the 2024 US presidential election. For period 2, relatively calmer than the rest of the sample, we expect a better fit of our volatility

FIGURE 3: Trading volume across Binance fee regime phases



models, as less extreme price movements are present. Going forward, the term *periods* will exclusively refer to the market-regimes described in Figure 2, while the word *phases* will strictly be used for the fee-regimes of Figure 3. Beyond these shared contextual periods (53% of Bitcoin-relevant events also affect Ethereum, and 72% of Ethereum-relevant events also affect Bitcoin), we verify whether this visual impression of joint price movement holds statistically using cointegration tests.

4.1.2 Cointegration Testing

Cointegration allows us to verify if two assets share a long-term equilibrium relation while individual prices remain non-stationary. If the Merge had a meaningful impact on Ethereum's price, this shift is expected to manifest as a break in the long-run equilibrium between the two assets. We use the Engle-Granger test (Engle and Granger, 1987) which regresses one price series on the other and tests the residuals for stationarity via an Augmented Dickey-Fuller test:

$$y_t = \alpha + \beta x_t + u_t \tag{9}$$

$$\Delta \hat{u}_t = \rho \hat{u}_{t-1} + \sum_{i=1}^{k} \phi_i \Delta \hat{u}_{t-i} + \varepsilon_t \tag{10}$$

where y_t and x_t represent the log-prices of the two assets, $\hat{u_t}$ is the estimated residual representing the long-term equilibrium error, ϕ_i are the autoregressive coefficients of the lagged residual differences and k is the lag length. The null hypothesis is $H_0 : \rho = 0$ (no cointegration). This approach depends on which variable is chosen as a regressor and only handles a single cointegration relation. We hence complement it using the Johansen trace test (Johansen, 1988), based on a vector error-correction representation, symmetric by construction:

$$\Delta Y_t = \Pi Y_{t-1} + \sum_{i=1}^{k-1} \Gamma_i \Delta Y_{t-i} + \varepsilon_t \tag{11}$$

where Y_t is an $n \times 1$ vector of endogenous non-stationary variables, Γ_i are short-run coefficient matrices, and $\Pi = \alpha\beta'$ with rank $(\Pi) = r$ gives the number of cointegrating vectors. α represents the adjustment speeds and β the cointegrating vectors. The presence of cointegration is tested via

$$\lambda_{trace}(r) = -T \sum_{i=r+1}^{n} \ln(1 - \hat{\lambda}_i) \tag{12}$$

where T is the sample size, n is the number of variables and $\hat{\lambda_i}$ the estimated eigenvalues of Π.

According to both tests, the full sample presents no significant cointegration despite a 0.81 correlation (Table 1). This suggests that this comovement doesn't reflect a stable long-run equilibrium relationship, that might be interrupted by shocks. This observation is evidenced in our event plots (Figure 2): despite over half of news being shared by both pairs, many events impacting Ethereum are missing from the

Bitcoin plot, and the opposite holds true. For subperiods, the first one shows significant cointegration according to Engle-Granger (5%), Johansen rank 0 (1%) tests, also highlighted by a 96% correlation. Period 2 shows an almost identical pattern, with significant Engle-Granger (1%) and Johansen rank 0 (5%) tests, and a 93% correlation. Period 3 presents no cointegration and a 0.62 correlation.

Since period 2 begins with the first Merge announcement, and assuming that investors anticipation implement this information ahead of the actual event, the results between period 1 and 2 indicate that no clear break in the Ethereum-Bitcoin long-run relationship is to be noticed around this event: cointegration and correlation are essentially unchanged. Moreover, the disappearance of cointegration in period 3 could be coherent with a "Bitcoin decoupling" hypothesis that can be drawn from events of the third period. The arrival of crypto-ETFs as early as January of 2024 for Bitcoin and only six months later (in July of 2024) for Ethereum might have led to a temporary mismatch in investment vehicles, as financial institutions active on traditional markets could take part in Bitcoin trading only, leading to diverging market dynamics relative to Ethereum.

Table 1: Engle-Granger and Johansen cointegration test results, by period (ETH-BTC)

Period	**Engle-Granger**	**Johansen (r≤0)**	**Johansen (r≤1)**	**Correlation**
Global	Not significant	Not significant	Not significant	0.81
Period 1	***	**	Not significant	0.96
Period 2	***	**	Not significant	0.93
Period 3	Not significant	Not significant	Not significant	0.62

4.1.3 Flight of Capital From PoW to PoS

We hypothesize that the Merge might have initiated, or been preceded by, a flight of capital from networks that underwent the PoW-to-PoS transition towards networks that remained PoW throughout this period. Ethereum falls in the former category while Bitcoin fits the latter, yet we also download Binance data for this period's top cryptocurrencies: BNB, XRP, DOGE, ADA, MATIC, DOT, LTC, SOL, SHIB, TRX, UNI, AVAX and BUSD. We obtain historical weekly market capitalization from CoinMarketCap's Cryptocurrency Historical Data Snapshot, ranging from July 3rd 2022 to November 27th 2022. We cover the entirety of the Merge period: before the first Merge announcement (2022-08-12), throughout, and after the Merge (2022-09-15). We also base our analysis on consensus mechanism evolution for these coins, obtained as of November 27th 2022 through Amoussou-Guenou et al. (2024) supplementary appendix.

A visual study of volume and price for coins with varying consensus mechanisms indicates no significant shift around the Merge period. Using a parallel axis plot that is not included here, we check for any relation between market capitalization and consensus mechanism throughout the Merge. Contrary to our hypothesis, no common trends are observed for coins that remained either PoW or PoS, nor for those that underwent the PoW-to-PoS transition. Coins seemingly all move together from one week to another independent of consensus mechanism, prompting us not to explore the dynamics of other coins any further.

4.2 HAR-RV Model

4.2.1 Model Design and Adaptation to Cryptocurrency Characteristics

To explore the dynamics linking volatility to volume, fees and liquidity during the periods surrounding the Merge, we use the Heterogeneous Autoregressive model of Realized Volatility (HAR-RV) introduced by Corsi (2009) to study how these variables affect model performances. HAR-RV is a three factor stochastic volatility model with a simple autoregressive structure based on measures of cascading volatilities realized over intervals of respectively one day, one week and one month. Its varying horizons represent fluctuating

expectations over different timeframes for next period's volatility. According to Andersen et al. (2007), HAR-RV has shown remarkably good volatility forecasting performance comparable to that of more complicated long-memory ARFIMA models. Realized volatility over a period of x trading days is defined as the average of daily quantities

$$RV_t^{(x)} = \frac{1}{x}\left(RV_t + RV_{t-1} + ... + RV_{t-x+1}\right) \tag{13}$$

where RV is the realized variance – also referred to as realized volatility – defined in Eq. 1. The empirical HAR-RV model is specified as follows:

$$RV_{t+1} = c + \beta^{(d)} RV_t^{(1)} + \beta^{(w)} RV_t^{(5)} + \beta^{(m)} RV_t^{(20)} + e_{t+1} \tag{14}$$

A benefit of having such a simple model is that it can easily be extended by adding additional regressors. Tseng et al. (2015) introduced lagged volume in the model, following the intuition of Clark (1973) who states that both traded volume and volatility are driven by the same underlying "news" variable and will therefore be positively correlated: this relationship has been widely documented for stock markets (Gallant et al., 1992; Andersen, 1996). We extend the model following this volume approach as well as using illiquidity metrics, cross-asset volatility and Binance's fees regime in order to observe how these features impact fit and forecasting performance. For example, an illiquid market is expected to be more volatile, with each order having a larger market-impact. Given the cointegration results discussed earlier, cross-asset volatility is expected to be significant between Ethereum and Bitcoin over the entire sample, but might deteriorate in period 3. Finally, the increased trading volume induced by the temporary removal of fees on Binance is expected to have a positive correlation with volatility, as it reduces trading friction and encourages more frequent trading. S&P 500 prices and risk-free rate were also regressed but were quickly excluded from the model, as index prices had no significant relation with cryptocurrency prices for most of our periods, while risk-free rate's slow evolution and high anticipation were too pronounced to add any meaningful improvement.

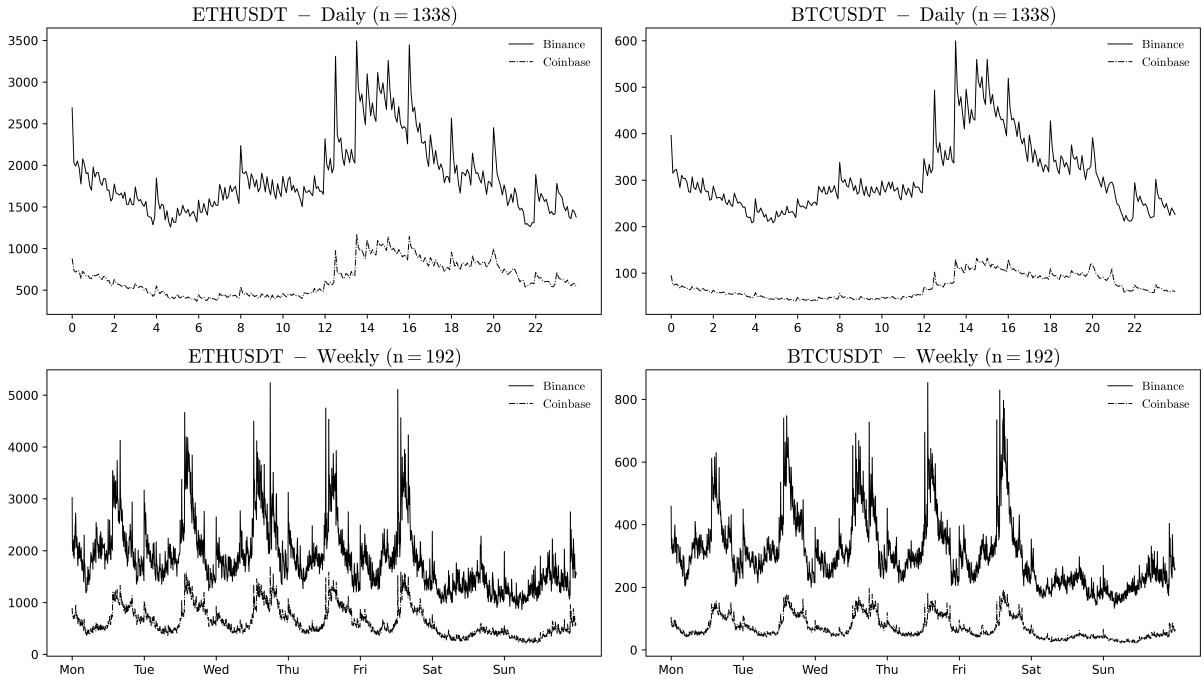
Unlike the 252 trading days observed on traditional financial markets, crypto trading happens live around the clock, with no weekend or holiday breaks. This characteristic prompts us to consider an adapted version of the HAR-RV model: instead of defining Eq. 13 over a week of five trading days and a month of twenty, we consider the case of live markets where a week consists of seven days and a month of thirty. However, Figure 4 shows that there is a clear intraday and weekly cycle of trading volume on Bitcoin and Ethereum pairs for both Binance and Coinbase data. UTC midday hours show higher volume, while weekends are characterized by limited activity. We hence prefer a classical 1-5-20 horizon configuration over a 1-7-30 model, because our data indicates that major cryptocurrencies' volume display the same investor attention span observed on traditional financial markets, consistent with cryptocurrency cycles described by Hansen et al. (2024).

4.2.2 Additive HAR-RV Specifications

We first add variables individually to the base HAR-RV model, in order to compare their fit over the entire sample to that of the initial model. Volume, Amihud ratio, realized Amihud as well as Kyle's lambda are each added as three cascading regressors based on daily, weekly and monthly averages, following the horizons of the original configuration. However, the cross-asset volatility, the jump component and the dummy variable are each a single addition to the model. The binary fee dummy is positive only during the fee-less phase, defined as:

$$fees_t = 1\{t \in [\text{2022-07-08}, \text{2023-03-22}]\} \tag{15}$$

FIGURE 4: Intraday and weekly cycle of trading volume (UTC)



The jump component jt follows Andersen et al. (2007), as

$$jt_t(\Delta) = \max\left(RV_t - BV_t, 0\right) \tag{16}$$

and adding it to the base model reproduces their HAR-RV-J specification. Table 2 reports adjusted R^2 of features added individually. Detailed coefficients are omitted for readability, but are analysed below.

For Ethereum, the base .188 adjusted R^2 shows meaningful but marginal improvement only with volume and Kyle's lambda, with individual fits of .206 and .191 respectively. Volume coefficients are positive but not significant with only the daily lag being significant (*). Kyle's lambda coefficients are negative on the daily (*) and weekly horizon, but positive for the monthly one (**). Cross-asset volatility, jump component, other liquidity measures and the fee dummy are not significant on their own and they worsen the fit. This is expected for the fee dummy, as Ethereum doesn't undergo any fee regime change during the period tackled by the dummy variable. For Bitcoin, the jump component and Kyle's lambda moderately improve the initial .130 fit, with respectively .133 and .135. Jump component is significant (-.4847**) when added to the base model. Volume and Amihud's ratio offer moderate improvement, while realized Amihud and the dummy leave the fit unchanged or slightly worse. The dummy coefficient is here insignificant when considered alone.

TABLE 2: Adj. R^2 of HAR-RV models with individual regressors

	HAR-RV	**Volume**	**Cross vol.**	**jt**	**Amihud**	**RAmihud**	**Kyle's λ**	**Fees**
ETH	.188	**.206**	.187	.187	.166	.187	**.191**	.187
BTC	.130	**.131**	.129	**.133**	**.131**	.130	**.135**	.130

Bold indicates improvements over the HAR-RV benchmark

Building on these results, we estimate additive models, summarized in Table 3 for the entire sample and in Appendix A.3 for periods. We consider a cumulative *base +volume +kyle +jt +cross_vol +fees* approach based on the improvements observed with individual variables.

For Ethereum over the full period, combined volume and Kyle's lambda improve fit with an .215

adjusted R^2, above that reached by the individual specifications, underlining that their respective effect remains when considered together. All further specifications marginally improve adjusted R^2 while deteriorating BIC: we hence retain the *volume +Kyle* specification for Ethereum over the full sample. Across periods, feature choice becomes regime dependent. Period 1 and 2 show similar HAR-RV fit, of .145 and .171 respectively. However, they diverge once additional regressors are introduced: period 1 penalizes complexity, with *+volume* being the best model (.166), while fit continues to improve with complexity in period 2, reaching .240 for *+volume +Kyle +jt*. Period 3 shows no exploitable fit (.015) with almost all coefficients being non-significant. These results suggest that additional regressors are only significant in period 2, while period 1 and 3 do not gain from added complexity.

For Bitcoin over the full period, the full specification *volume +Kyle +jump component +cross_volatility +fee* has the best performance for both R^2 and AIC. The dummy's coefficient (-.0008***) confirms its explanatory role once other factors are taken into account. The negative sign of the coefficient indicates that the fee-less period is associated with lower volatility, the opposite of our initial hypothesis. This result could suggest that the increased volume associated with the fee removal reflects a more liquid market, where higher volume smooths out otherwise erratic activity. Period analysis analysis a pattern quite similar to that of Ethereum: the first period favors the full specification (.108), the second favors the *+volume +Kyle +jt* model (.109), and period 3 shows negligible explanatory power, with the base model being preferred by AIC and BIC.

Overall, past volatility and trading volume drive most of the explanatory power for both assets, as highlighted by Tseng et al. (2015). Microstructure metrics (Kyle's lambda, jump component) add marginal improvement, while cross-asset volatility spillovers and fee-regime effects matter only when controlling for other variables. The higher fit for the second period noticed for both pairs suggests that volatility dynamics are more predictable by our regression in low-volatility episodes. These findings can be compared to the forecasting performance of these models, explored in the next section.

Since the first Merge announcement falls at the start of period 2, these results also speak to our original question. If the consensus mechanism shift had impacted the mechanisms driving Ethereum's volatility, we would expect the explanatory power and the coefficients of our added regressors to shift between period 1 and 2 for Ethereum specifically, beyond what is observed for Bitcoin as a counterfactual. Instead, both cryptocurrencies show a similar improvement of fit over period 2. This suggests that this improved fit reflects the overall calmer market regime described in Section 4 and that the drivers of Ethereum's volatility were not altered by the Merge, at least not when considering the variables we added as regressors.

4.2.3 Proxy and Loss Functions for Volatility Forecasting

The evaluation of volatility forecasts is hindered by the aforementioned fact that the variable of interest – intraday volatility – cannot be directly measured, but only approximated. Patton (2011) shows that the choice of volatility proxy matters in this case, as certain loss functions can lead to distorted ranking of competing forecasts. He derives conditions for a loss function to be robust to such distortion and shows that only mean squared-error (MSE, Eq. 17) and QLIKE (Eq. 18) loss functions satisfy them.

$$L_{MSE}(\hat{\sigma}^2, h) = (\hat{\sigma}^2 - h)^2 \tag{17}$$

$$L_{QLIKE}(\hat{\sigma}^2, h) = \log h + \frac{\hat{\sigma}^2}{h} \tag{18}$$

where $\hat{\sigma^2}$ denotes the volatility proxy and h is the conditional variance forecast.

Between those candidates, we select QLIKE over MSE for two reasons presented by Patton (2011). First, Patton and Sheppard (2009) shows that Diebold-Mariano-West tests of equal predictive accuracy – tests that verify whether one forecast is on average significantly more accurate than another – have higher power when using QLIKE loss instead of MSE loss. Second, since QLIKE uses standardized error

TABLE 3: HAR-RV models estimates with additive regressors (full sample)

A. ETHUSDT (n=1317)

	base	**+volume**	**+kyle**	**+jt**	**+cross_vol**	**+fees**
const	.0007***	-.0096***	-.0115***	-.0116***	-.0116***	-.0129***
RV_1	.2009***	.1428***	.1730***	.1967***	.2716***	.2821***
RV_5	.1575***	.1493***	.1567***	.1462***	.1402***	.1415***
RV_{20}	.1032***	.0378	-.0893*	-.0883*	-.0891*	-.0964*
vol_1		.0003*	.0003*	.0003*	.0004**	.0004**
vol_5		.0002	.0004	.0004	.0004	.0004
vol_{20}		.0003	.0002	.0002	.0002	.0003
$kyle_1$			-.1590	-.1652	-.1558	-.1530
$kyle_5$			-.3751*	-.3704*	-.3736*	-.3851*
$kyle_{20}$			.6915***	.6860***	.6936***	.6730***
jt				-.2429	-.2350	-.2203
cross_vol					-.1391	-.1652
fees						-.0002
R^2	.1896	.2097	.2206	.2214	.2223	.2237
Adj. R^2	.1878	.2061	.2152	.2154	.2157	.2165
AIC	-12639.57	-12666.61	-12678.93	-12678.29	-12677.80	-12678.17
BIC	-12618.83	-12630.32	-12627.10	-12621.28	-12615.60	-12610.79

B. BTCUSDT (n=1317)

	base	**+volume**	**+kyle**	**+jt**	**+cross_vol**	**+fees**
const	.0005***	.0004	.0003	.0002	.0002	-.0035***
RV_1	.1565***	.1454***	.1730***	.2259***	.2139**	.1102
RV_5	.1419***	.0945*	.0694	.0488	.0473	.0154
RV_{20}	.1512***	.1950***	.1327***	.1401***	.1401***	.0923*
vol_1		.0000	.0000	.0000	.0000	.0001
vol_5		.0004*	.0004**	.0004**	.0004**	.0006***
vol_{20}		-.0004**	-.0005***	-.0005***	-.0005***	-.0004**
$kyle_1$			-.0007	-.0007*	-.0007*	-.0006
$kyle_5$			-.0000	.0001	.0001	-.0001
$kyle_{20}$			.0015**	.0014**	.0014**	.0014**
jt				-.4958**	-.4878**	-.4233**
cross_vol					.0072	.0557
fees						-.0008***
R^2	.1318	.1355	.1441	.1483	.1483	.1567
Adj. R^2	.1298	.1315	.1382	.1417	.1411	.1490
AIC	-13867.28	-13866.90	-13874.03	-13878.49	-13876.51	-13887.66
BIC	-13846.55	-13830.62	-13822.19	-13821.47	-13814.32	-13820.28

* $p < .1$, ** $p < .05$, *** $p < .01$

TABLE 4: Out-of-sample QLIKE forecast losses by model and period

A. ETHUSDT

	Period			
Model	*Full*	*1*	*2*	*3*
HAR-RV	**-5.7029**	-4.6856	-6.0968	-5.4564
+volume	-5.5225	-4.8247	-5.9600	-5.5136
+Kyle	-5.6794	-4.7026	-6.0839	-5.4509
+cross_vol	-5.6991	-4.6655	-6.0925	-5.4559
+jt	-5.7009	-4.7199	-6.1101	-5.4644
volume	-5.4884	**-4.8537**	-5.9504	**-5.5673**
Kyle	-5.6850	-4.3878	**-6.1183**	-5.3841

B. BTCUSDT

	Period			
Model	*Full*	*1*	*2*	*3*
HAR-RV	-6.1551	-5.3999	-6.0554	-6.0362
+volume	-6.1519	**-5.4447**	-5.8261	-5.9833
+Kyle	-6.1226	-5.3878	-5.8947	-6.0261
+cross_vol	-6.1519	-5.3563	-6.0513	-6.0247
+jt	-6.1588	-5.3948	**-6.0606**	-6.0386
volume	**-6.1670**	-5.4144	-5.7494	**-6.0658**
Kyle	-6.1353	-5.2680	-5.9087	-6.0209

Bold indicates best, underlined indicates second best.

$\hat{\sigma}^2/h$ rather than raw error $\hat{\sigma}^2 - h$, it is less sensitive to extreme observations, which is relevant in our application since heterogeneous volatility regimes are observed depending on period. Our forecast results are therefore presented using the QLIKE loss function. As for our volatility proxy, we retain realized volatility (Eq. 1) rather than other proxies such as bipower variation (Eq. 2): in our setting, the jump component (Eq. 16) implies that BV would be a biased proxy whenever jumps are present, and would hence violate the A1 assumption required for Patton's robustness result. Moreover, our specifications were designed to predict RV_{t+1} along standard HAR-RV recommendations (Corsi, 2009), hence why RV is the natural choice for out-of-sample evaluation: using BV would lead to a mismatch between estimation and evaluation targets that could differ in scale.

For each specification, we compute one-step-ahead out-of-sample forecasts of RV_{t+1}. We use an expanding window to estimate our coefficients: starting from an initial training set covering 80% of the relevant chronological period, the model is re-estimated at each step t using all data available up to t. The window is then extended by one observation and the procedure is repeated for the remaining 20% of the period: our out-of-sample data. We then compute the QLIKE loss function for each forecast and report the average over the entire period. We experiment with the base HAR-RV model, as well as both the individual and additive models presented earlier. Additive models that performed well in-sample aren't presented here as they had disappointing out-of-sample performance, a striking example of overfitting. We also experiment with models regressing volume and Kyle's lambda alone on realized volatility, without the base HAR-RV component, although we follow the cascading structure over daily, weekly and monthly averages of Eq. 14. Results are reported in Table 4.

For Ethereum over the full period, base HAR-RV is the best model. All additions of individual variables worsen it, contrary to the improvement of individual variables found in Table 2. For underlying periods, only volume and the jump component improve the model with best performing models for period 1, 2 and 3 respectively being *+volume*, *+jt* and *+volume*. However, for the first and last period, regressing

cascading volume alone yields improved forecasting capacities compared to each period's best model. For period 2, regressing Kyle's lambda alone gives a lower (better) QLIKE than *+jt*, suggesting that this measure of illiquidity is a better indicator of RV in periods of low volatility, as trade-level data carries more information than aggregated, small-scale returns.

For Bitcoin, only the jump component improves the base model over the full period. For each period, volume and the jump component are the main drivers of improvement, *+volume* for the first period and *+jt* for the second and last. Regressing volume alone still yields high results for period 1 and 3, even higher than those obtained when regressing HAR-RV with added volume cascading features, underlining how regressing volatility worsens forecasting performance for this specific period. In period 3 specifically, HAR-RV alone gives a QLIKE of -6.0362, while HAR-RV+volume gives a higher (worse) -5.9833, and volume alone gives a lower (better) -6.0658. This time, Kyle's lambda alone worsens performance for period 2.

Volume's stronger performance when regressed alone in contexts of higher volatility (first and last periods) confirms its enhanced explanatory power for periods of higher volatility, when its cascading horizons are taken into account. This is consistent with the volume-volatility relationship discussed in Section 4.2.1. However, Kyle's lambda forecasting capacities for volatility are more mixed: no conclusion can be drawn from its varying performance across assets during periods of low volatility. More broadly, volume and Kyle's lambda improved in-sample fit when added (Table 3) but worsen out-of-sample forecasts for the full sample of both assets (Table 4). This discrepancy suggests these regressors may capture sample-specific patterns that are not generalizable. Consistent with the in-sample results of Section 4.2.2, forecastingc performance shows no distinct pattern for Ethereum around the Merge beyond what is observed for Bitcoin. This once again reinforces our conclusion: the consensus mechanism evolution left the drivers of volatility unchanged.

5 Conclusion

This report explores whether Ethereum's transition from PoW to PoS had an impact on spot trading dynamics, using Bitcoin as a counterfactual. The Merge itself is found to be largely irrelevant to price and volatility dynamics: no clear related effect is identified across event analysis, cointegration testing, PoW-to-PoS capital flight hypothesis and HAR-RV volatility modelling. On the other hand, Binance's temporary removal of trading fees – a promotional stunt that happened shortly after the Merge – impacted Bitcoin's trading metrics – notably volume – more than the protocol upgrade did for those of Ethereum. This is a reassuring finding for the Ethereum network: the Merge led to an energy-efficient and scalable protocol, safer according to John et al. (2025), without any adverse impact on trading activity.

We also find that Ethereum-Bitcoin cointegration is unstable over periods: present only before the 2024 introduction of Bitcoin spot ETFs, it suggests a decoupling between the two cryptocurrencies after this event. A basic visual analysis shows no evidence of capital flight between PoW and PoS cryptocurrencies around the Merge. In the HAR-RV model, lagged volume and Kyle's lambda are the strongest individual regressors for both assets. Volume outperforms the full additive specifications when regressed alone against realized volatility, but only in periods of higher volatility. On the data side, our day-by-day streaming approach allowed us to use billions of aggregated trades needed for Kyle's lambda computation, and our bin reconstruction fits counterfactual Coinbase prices.

This remains an exploratory analysis: we do not estimate a difference-in-difference model and do not explore derivatives data, which is why our conclusions are descriptive and further modelling would be needed for causal results.

A natural next step would hence be a formal DiD design, isolating the Merge's effect on Ethereum. This raises the question of an appropriate cutoff date, given the progressive integration of information into metrics, between rumors, Merge announcement and the execution itself. Potential spill-over effects onto

Bitcoin would need to be addressed as well. Such a design would require exchange data free of Bitcoin's fee-shift observed on Binance. Granularity wouldn't be a hurdle, as 5m candlesticks appear sufficient as long as Kyle's lambda and other trade-level indicators aren't included in the regression. A further analysis of cryptocurrency derivatives markets would also be valuable, as leverage and anticipation in pricing may lead to markets reacting more abruptly to a protocol shift than spot trading does. Finally, we only used illiquidity metrics in a visual analysis and as regressors for volatility, not as a variable that could be explained by other metrics.

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A Appendix

A.1 Events Timeline

List of events linked to main volume spikes for ETHUSDT and BTCUSDT (2021-05 to 2024-12)

2021

1. 2021-05-19 – *Major crypto market crash amplified by high leverage and cascading liquidations*
2. 2021-06-22 – *China steps up efforts against Bitcoin*
3. 2021-07-26 – *Amazon's Bitcoin rumors skyrocket prices, liquidating $1B in short BTC positions*
4. 2021-08-05 – *Ethereum network; London hard fork*
5. 2021-09-07 – *Bitcoin becomes legal currency in El Salvador*
6. 2021-12-04 – *Spot selling drives price drop, triggering a long squeeze in the derivatives market*

2022

7. 2022-01-21 – *Anticipation of Fed rates hike*
8. 2022-02-24 – *Market-wide selloff following Ukraine invasion*
9. 2022-05-09 – *Beginning of the LUNA/UST depeg and Terra collapse*
10. 2022-05-12 – *LUNA/UST depeg, Terra ecosystem collapse*
11. 2022-06-13 – *Celsius blocks customer withdrawals*
12. 2022-07-08 – *Binance launches zero fee bitcoin trading*
13. 2022-08-12 – *Ethereum network; Mainnet Merge incoming (blog announcement)*
14. 2022-08-24 – *Ethereum network; Mainnet Merge Announcement (official announcement)*
15. 2022-09-13 – *US inflation higher than expected, reinforcing fear of Fed rates hike*
16. 2022-09-15 – *Ethereum network; Paris upgrade (the Merge)*
17. 2022-09-27 – *Bank of England stabilizing UK bond market fuels hopes of eased rates hike*
18. 2022-11-08 – *FTX halts withdrawals, Binance announces then cancels FTX acquisition*

2023

19. 2023-01-12 – *US slowing inflation drives prices up*
20. 2023-03-14 – *Fed protecting SVB and Signature Bank deposits eases banking contagion fears*
21. 2023-03-22 – *Binance stops zero fee bitcoin trading*
22. 2023-06-21 – *BlackRock's Bitcoin ETF sparks institutional buying*
23. 2023-08-17 – *Crypto sinks on rising US bond yields*
24. 2023-10-24 – *Growing anticipation of imminent BlackRock spot ETF approval*
25. 2023-11-09 – *Rally fueled by anticipation of spot Bitcoin ETF approval*

2024

26. 2024-01-10 – *SEC approves first US spot Bitcoin ETFs*
27. 2024-01-10 – *Inflows into new spot Bitcoin ETFs outpace Bitcoin supply*
28. 2024-03-05 – *Bitcoin reaches ATH, followed by profit-taking*
29. 2024-03-20 – *Rates are maintained but rate cuts are projected*
30. 2024-04-13 – *Iranian drone attack on Israel*
31. 2024-05-23 – *SEC approves filing for spot Ethereum ETFs*
32. 2024-07-05 – *Mt. Gox exchange starts repaying $9B in Bitcoin to creditors*
33. 2024-08-05 – *Unwinding of the yen carry trade: VIX spike, Ethereum sales by Jump Trading*
34. 2024-11-06 – *Trump's election sparks a crypto rally*
35. 2024-11-11 – *Market rally pricing friendlier Trump administration after elections win*
36. 2024-12-05 – *Bitcoin crosses $100,000 after Trump names Paul Atkins as next SEC chair*

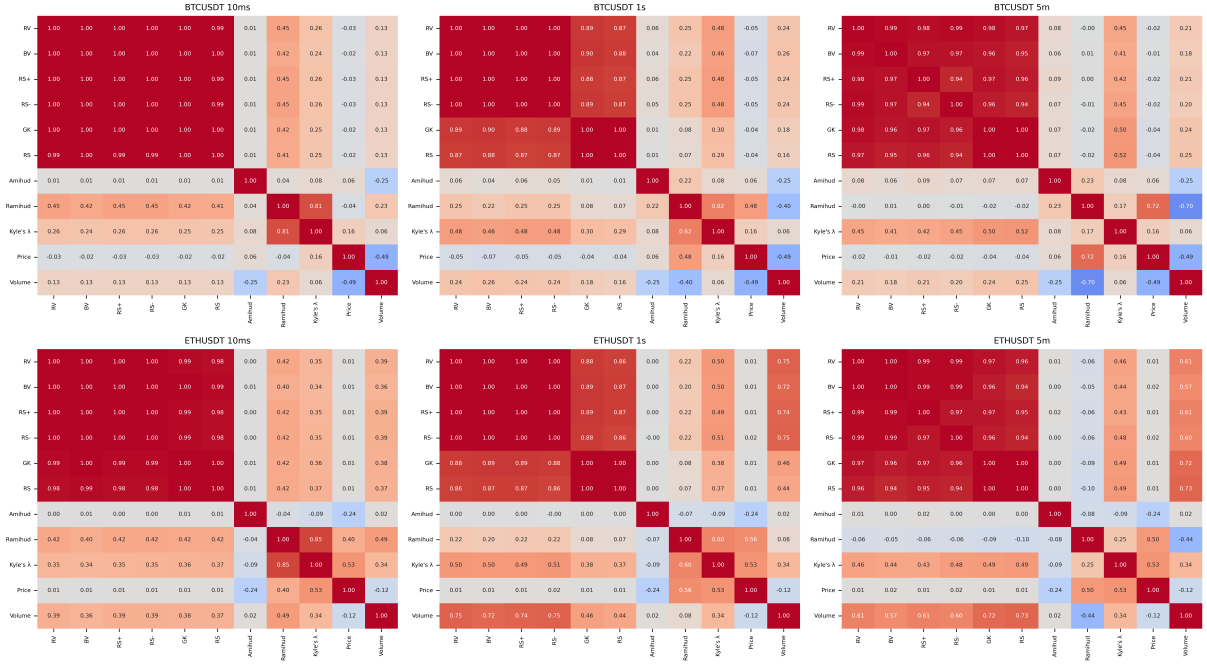
A.2 Correlation Analysis

In Figure 5, daily metrics correlations are measured over 1338 days, with each day's estimate built from multiple intraday bin sizes (10ms, 1s and 5m).

Across both assets, realized and range-based volatility estimators converge at 10ms. They then split into trade-based and range-based families capturing distinct volatility components at 1s, before reconverging at 5m. This behavior is consistent with high frequency noise imposing a common variance independent of estimator type (Hansen and Lunde, 2006; Bandi and Russell, 2008). Once this noise disappears, a slight upside/downside asymmetry is observed for BTCUSDT only. Volatility and liquidity correlation reverses as scale increases, from correlation with realized Amihud at 10ms bins to correlation with Kyle's lambda at 5m bins, while Amihud remains rather independent. Relationship of volatility with price is nonexistent at every scale, and that with volume peaks at 1s.

Liquidity measures show a similar granularity dependence, varying depending on the asset. For BTCUSDT, correlation between Amihud and realized Amihud strengthens as bin size increases. Correlation of Amihud and volatility remains negative for all illiquidity measures for ETHUSDT. For both assets, Kyle's lambda correlation moves closer to that of realized Amihud as bin size reduces, suggesting that price impact and realized illiquidity reflect the same phenomenon at the trade level but diverge over longer intervals. As expected since they are both computed independently of bin size, Amihud and Kyle's lambda have a stable correlation with price and volume, while Realized Amihud becomes negatively correlated with volume as bin size increases.

FIGURE 5: Correlation heatmap of volatility and liquidity metrics across intraday frequencies



A.3 Additive Models by Period

TABLE 5: Additive HAR-RV models by Period

A. ETHUSDT, Period 1 (n=446)

	base	**+volume**	**+kyle**	**+jt**	**+cross_vol**	**+fees**
const	.0015***	-.0086*	-.0089	-.0087	-.0093	-.0130
RV_1	.1523***	.0436	.0778	.0724	.1783	.1991
RV_5	.1761***	.1457**	.1702**	.1729**	.1663**	.1677**
RV_{20}	.0364	.0549	-.0346	-.0336	-.0374	-.0611
vol_1		.0010**	.0009*	.0009*	.0010*	.0011**
vol_5		.0011	.0011	.0011	.0011	.0010
vol_{20}		-.0013*	-.0012	-.0012	-.0012	-.0010
$kyle_1$			-.1221	-.1156	-.0908	-.0825
$kyle_5$			-.4286	-.4299	-.4487	-.4581
$kyle_{20}$			.5236	.5202	.5380	.5399
jt				.0991	.0728	.0683
cross_vol					-.2050	-.2516
fees						-.0006
R^2	.1498	.1774	.1839	.1840	.1858	.1876
Adj. R^2	.1441	.1662	.1671	.1652	.1651	.1650
AIC	-3993.45	-4002.14	-3999.70	-3997.74	-3996.71	-3995.69
BIC	-3977.05	-3973.44	-3958.69	-3952.64	-3947.50	-3942.38

B. ETHUSDT, Period 2 (n=517)

	base	**+volume**	**+kyle**	**+jt**	**+cross_vol**	**+fees**
const	.0004***	-.0083***	-.0116***	-.0097***	-.0096***	-.0070**
RV_1	.3646***	.3671***	.3521***	.5651***	.5918***	.5499***
RV_5	-.0091	-.0128	.0850	.0353	.0311	.0407
RV_{20}	.2108**	-.1182	-.1385	-.1112	-.1099	-.1429
vol_1		-.0001	-.0000	-.0001	-.0001	-.0002
vol_5		.0000	-.0001	.0001	.0001	-.0001
vol_{20}		.0007**	.0012***	.0010***	.0010**	.0009**
$kyle_1$			-.1775	-.0067	-.0004	.0113
$kyle_5$			-.6266	-.7114	-.7004	-.6188
$kyle_{20}$			-1.7171**	-1.5849**	-1.5897**	-2.0007***
jt				-1.1698***	-1.1562***	-1.1915***
cross_vol					-.0491	.0221
fees						.0004***
R^2	.1711	.1909	.2219	.2552	.2554	.2673
Adj. R^2	.1662	.1814	.2081	.2405	.2392	.2499
AIC	-5390.93	-5397.43	-5411.64	-5432.29	-5430.42	-5436.75
BIC	-5373.94	-5367.70	-5369.16	-5385.56	-5379.44	-5381.53

C. ETHUSDT, Period 3 (n=356)

	base	**+volume**	**+kyle**	**+jt**	**+cross_vol**	**+fees**
const	.0009***	-.0065	-.0062	-.0053	-.0054	-.0054
RV_1	.0921	.0530	.0454	.1423	.2233	.2233
RV_5	.1430	.1600	.1583	.1470	.1457	.1457
RV_{20}	-.0197	-.2116	-.1756	-.1861	-.1870	-.1870
vol_1		.0002	.0002	.0001	.0001	.0001
vol_5		-.0001	-.0002	-.0002	-.0002	-.0002
vol_{20}		.0006	.0006	.0006	.0006	.0006
$kyle_1$			.0277	.0426	.0405	.0405
$kyle_5$			.0323	.0075	.0113	.0113
$kyle_{20}$			-.1304	-.1204	-.1091	-.1091
jt				-.4624	-.5187	-.5187
cross_vol					-.1270	-.1270
fees						.0000
R^2	.0235	.0345	.0350	.0364	.0371	.0371
Adj. R^2	.0152	.0179	.0099	.0085	.0064	.0064
AIC	-3606.97	-3604.99	-3599.20	-3597.70	-3595.98	-3595.98
BIC	-3591.47	-3577.87	-3560.45	-3555.08	-3549.48	-3549.48

D. BTCUSDT, Period 1 (n=446)

	base	**+volume**	**+kyle**	**+jt**	**+cross_vol**	**+fees**
const	.0010***	.0012	.0023	.0024	.0024	-.0086*
RV_1	.1283**	.0565	.0864	.1299	.1263	-.0488
RV_5	.1292*	.0538	.0599	.0396	.0391	-.0162
RV_{20}	.0662	.1734**	.1602*	.1781**	.1782**	.0553
vol_1		.0004	.0003	.0004	.0004	.0006*
vol_5		.0009*	.0010*	.0009*	.0009*	.0014**
vol_{20}		-.0013**	-.0014***	-.0015***	-.0015***	-.0011**
$kyle_1$			-.0005	-.0005	-.0005	-.0003
$kyle_5$			-.0008	-.0007	-.0007	-.0011
$kyle_{20}$			.0007	.0005	.0005	.0013
jt				-.6439*	-.6422*	-.5670
cross_vol					.0022	.0808
fees						-.0016***
R^2	.0825	.1003	.1082	.1143	.1143	.1324
Adj. R^2	.0763	.0880	.0898	.0939	.0918	.1083
AIC	-4432.21	-4434.94	-4432.87	-4433.92	-4431.92	-4439.12
BIC	-4415.81	-4406.24	-4391.87	-4388.82	-4382.72	-4385.81

E. BTCUSDT, Period 2 (n=517)

	base	**+volume**	**+kyle**	**+jt**	**+cross_vol**	**+fees**
const	.0004***	-.0009*	.0003	-.0000	.0000	-.0001
RV_1	.2171***	.2370***	.2562***	.5433***	.4906***	.4855***
RV_5	.1371	.0800	.0745	.0193	.0110	.0089
RV_{20}	-.0749	-.2044	-.3130*	-.3335*	-.3220*	-.3239*
vol_1		-.0001	-.0001	-.0001	-.0001	-.0001
vol_5		.0002	.0001	.0002	.0001	.0001
vol_{20}		.0000	-.0000	.0000	.0000	.0000
$kyle_1$			-.0007	-.0016	-.0015	-.0014
$kyle_5$			.0007	.0011	.0013	.0013
$kyle_{20}$			.0027	.0021	.0019	.0019
jt				-1.2135***	-1.1760***	-1.1700***
cross_vol					.0331	.0353
fees						-.0000
R^2	.0719	.0873	.0929	.1266	.1270	.1270
Adj. R^2	.0665	.0765	.0768	.1093	.1079	.1062
AIC	-5749.97	-5752.59	-5749.80	-5767.37	-5765.59	-5763.60
BIC	-5732.98	-5722.86	-5707.32	-5720.64	-5714.61	-5708.37

F. BTCUSDT, Period 3 (n=356)

	base	**+volume**	**+kyle**	**+jt**	**+cross_vol**	**+fees**
const	.0005***	-.0037	-.0029	-.0028	-.0027	-.0027
RV_1	.0491	.0437	.0783	.0675	.1344	.1344
RV_5	.2114*	.3102*	.1585	.2130	.2156	.2156
RV_{20}	.0863	-.2522	-.2835	-.2893	-.2921	-.2921
vol_1		.0000	.0000	.0001	.0000	.0000
vol_5		-.0003	-.0003	-.0003	-.0004	-.0004
vol_{20}		.0007*	.0006	.0006	.0006	.0006
$kyle_1$			-.0005	-.0003	-.0003	-.0003
$kyle_5$			.0019	.0017	.0017	.0017
$kyle_{20}$			.0002	.0002	.0002	.0002
jt				-.8907	-.9559	-.9559
cross_vol					-.0387	-.0387
fees						.0000
R^2	.0333	.0433	.0517	.0563	.0568	.0568
Adj. R^2	.0250	.0269	.0271	.0290	.0266	.0266
AIC	-3963.04	-3960.77	-3957.91	-3957.63	-3955.80	-3955.80
BIC	-3947.54	-3933.65	-3919.16	-3915.01	-3909.30	-3909.30

* $p < .1$, ** $p < .05$, *** $p < .01$